ATOM COUNT CARD



A. PRODUCT IDENTITY

- PRODUCT: NESCAFÉ CLASSIC (100% PURE INSTANT BLACK COFFEE)
- COMPOUND CHOSEN: CAFFEINE ($C_8H_{10}N_4O_2$)
- Printed Label Value / Basis: 2.0g of pure soluble coffee granules
- Why it matters: It is teh fuel of every engineering major so they could function well during their classes
- Student: Jamil Benito B. Inocencio

B. BASE QUANTITATIVE DATA

Quantity	Value	Unit	Source	Precision / Sig. Figs.
Serving Size (Net Content)	2.0	g	Product Label	2 s.f.
Caffeine per Serving	60	mg	Nestlé Product Spec	2 s.f (6.0×10^1 mg)
Chemical Formula	$C_8H_{10}N_4O_2$	-	PubChem	Exact Subscripts
Atomic Mass (Carbon, C)	12.011	g/mol	CIAAW / IUPAC	Not limiting (3 decimals)
Atomic Mass (Hydrogen, H)	1.008	g/mol	CIAAW / IUPAC	Not limiting (3 decimals)
Atomic Mass (Nitrogen, N)	14.007	g/mol	CIAAW / IUPAC	Not limiting (3 decimals)
Atomic Mass (Oxygen, O)	15.999	g/mol	CIAAW / IUPAC	Not limiting (3 decimals)
Avogadro Constant (N_A)	$6.02214076 \times 10^{23}$	mol^{-1}	BIPM (2019)	Exact

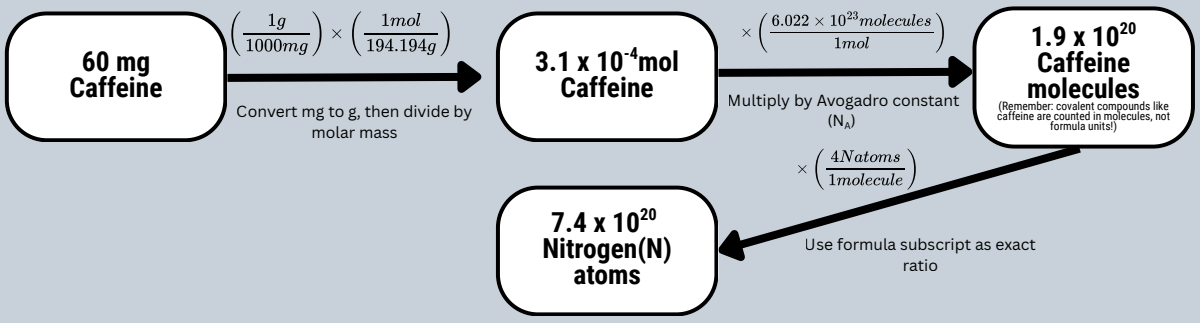
Assumption Statement: "1 sachet (2.0g) contains 60 mg of pure caffeine."

C. MOLAR MASS-BUILD UP

$M(C_8H_{10}N_4O_2) = 8(12.011) + 10(1.008) + 4(14.007) + 2(15.999)$
$= 96.088 + 10.080 + 56.028 + 31.998$
$= 194.194$ g/mol

Element	Atoms	Atomic Mass (g/mol)	Contribution (g/mol)	% by Mass
C	8	12.011	96.088	49.480%
H	10	1.008	10.080	5.191%
N	4	14.007	56.028	28.852%
O	2	15.999	31.998	16.477%
Total	24	-	194.194	100.000%

D. THE MOLE MAP



E. INSIDE THE FORMULA

CALCULATION 1: MOLAR MASS BUILD-UP
M = 8(12.011) + 10(1.008) + 4(14.007) + 2(15.999) = 194.194g/mol Caffeine

CALCULATION 2: LABEL BRIDGE (MASS OF CAFFEINE IN ONE SACHET)
60 mg caffeine x (1 g / 1000 mg) = 0.060 g = 6.0 x 10^{-2}g Caffeine

CALCULATION 3: MASS TO MOLES
6.0 x 10^{-2} g Caffeine x (1 mol Caffeine / 194.194 g Caffeine) = 3.0897 x 10^{-4}mol = 3.1 x 10^{-4} mol Caffeine

CALCULATION 4: MOLES TO MOLECULES
3.0897 x 10^{-4} mol x (6.02214076 x 10^{23} molecules / 1 mol) = 1.8606 x 10^{20} Caffeine molecules

CALCULATION 5: ATOMS OF NITROGEN INSIDE 1 SACHET
1.8606 x 10^{20} molecules x (4 N atoms / 1 molecule) = 7.4425 x 10^{20} Nitrogen Atoms

Calculation 6: Percent Composition (Nitrogen) & Two-Route Check

$$\%N = \left[\frac{4\,(14.007)}{194.194}\right] \times 100\% = 28.852\%$$

Route A: 0.060 g Caffeine x 0.28852 = 0.017311 g = 1.7 x 10^{-2}g N
Route B:

$$3.0897 \times 10^{-4} molCaffeine \times \left(\frac{4molN}{1molCaffeine}\right) \times \frac{14.007gN}{1molN} = 0.017311g = 1.7 \times 10^{-2} gN$$

Calculation 7: Reverse (% Composition to Empirical & Molecular Formula)
In 100g sample of pure caffeine:
C: 49.480g / 12.011 = 4.1196 mol / 1.0299 = 4.000
H: 5.191 g / 1.008 = 5.1498 mol / 1.0299 = 5.000
N: 28.852 g / 14.007 = 2.0598 / 1.0299 = 2.000
O: 16.477 g / 15.999 = 1.0299 mol / 1.0299 = 1.000
Empirical Formula = $C_4H_5N_2O$

$$n = \frac{M_{molecular}}{M_{empirical}} = \frac{\frac{194.194g}{mol}}{\frac{97.097g}{mol}} = 2.0000 \rightarrow C_8H_{10}N_4O_2$$

F. SCALE UP

Calculation 8: Scale-up & Reverse
Scale-up (1 class of 40 students drinking 1 sachet per day for 5 days = 200 sachets):
200 sachets x (3.0897 x 10^{-4}mol/sachet) = 0.06179 mol = 6.2 x 10^{-2} mol Caffeine

Reverse (How many sachets contain 1.00 mol Caffeine?):

1.00 mol Caffeine x (1 sachet / 3.0897 x 10^{-4} mol) = 3236.5 = 3.2 x 10^3 sachets (or cups)

G. SIGNIFICANT FIGURE CHECK

Calculator value: 1.860648834 x 10^{20} Caffeine molecules
(or 7.442595336 x 10^{20} Nitrogen atoms)
Reported value: 1.9 x 10^{20} Caffeine molecules (or 7.4 x 10^{20} Nitrogen atoms)
Limiting value: 60 mg (or 2.0 g) from the label, which has 2 significant figures.
Exact Constants(Do not limit precision):
Avogadro constant (N_A = 6.02214076 x 10^{23})
Subscripts in formula ($C_8H_{10}N_4O_2$)
Unit conversion factors (1 g = 1000 mg)

ONE-NUMBER TAKEAWAY:

"A single 2.0g sachet of Nescafé Classic contains roughly 1.9x10^{20} caffeine molecules, yet it would take over 3,200 sachets to collect just 1 mole of caffeine, demonstrating how impossibly large Avogadro's number really is!"

H. REFERENCES AND AI DISCLOSURE

- BUREAU INTERNATIONAL DES POIDS ET MESURES. (2019). THE INTERNATIONAL SYSTEM OF UNITS (SI) (9TH ED.). HTTPS://WWW.BIPM.ORG/EN/PUBLICATIONS/SI-BROCHURE
- COMMISSION ON ISOTOPIC ABUNDANCES AND ATOMIC WEIGHTS. (2024). ABRIDGED STANDARD ATOMIC WEIGHTS. INTERNATIONAL UNION OF PURE AND APPLIED CHEMISTRY. HTTPS://WWW.CIAAW.ORG/ABRIDGED-ATOMIC-WEIGHTS.HTM
- NATIONAL CENTER FOR BIOTECHNOLOGY INFORMATION. (N.D.). PUBCHEM COMPOUND SUMMARY FOR CID 2519, CAFFEINE. RETRIEVED OCTOBER 28, 2026, FROM HTTPS://PUBCHEM.NCBI.NLM.NIH.GOV/COMPOUND/CAFFEINE
- NESTLÉ PHILIPPINES. (N.D.). NESCAFÉ CLASSIC 100% PURE SOLUBLE COFFEE, 2 G SACHET [PRODUCT LABEL]. PHOTOGRAPH BY INOCENCIO, OCTOBER 28, 2026.

"AI (GEMINI) WAS USED TO HELP REVIEW CALCULATION STEPS, CHECK SIGNIFICANT FIGURE RULES, AND FORMAT THE LAYOUT OF THIS INFOGRAPHIC. ALL LABEL VALUES, ATOMIC MASSES, CALCULATIONS, AND REFERENCES WERE INDEPENDENTLY VERIFIED BY THE STUDENT."